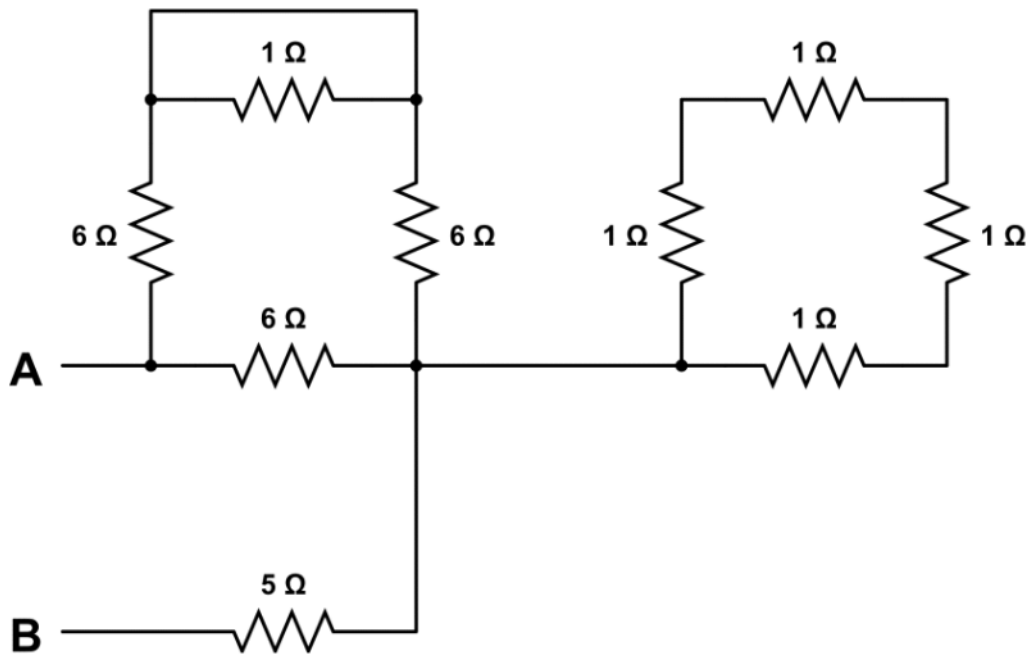


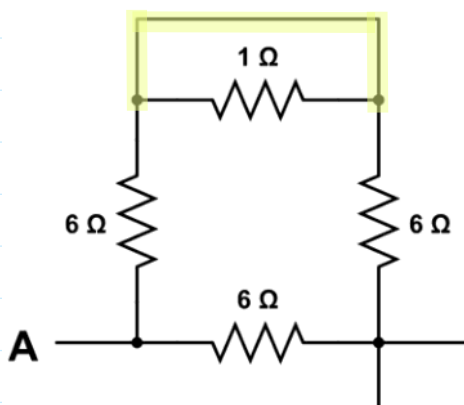
Practice Quiz 2 Solution

Wednesday, September 10, 2025 1:12 PM

For the circuit shown below, find the equivalent resistance seen between points A and B.

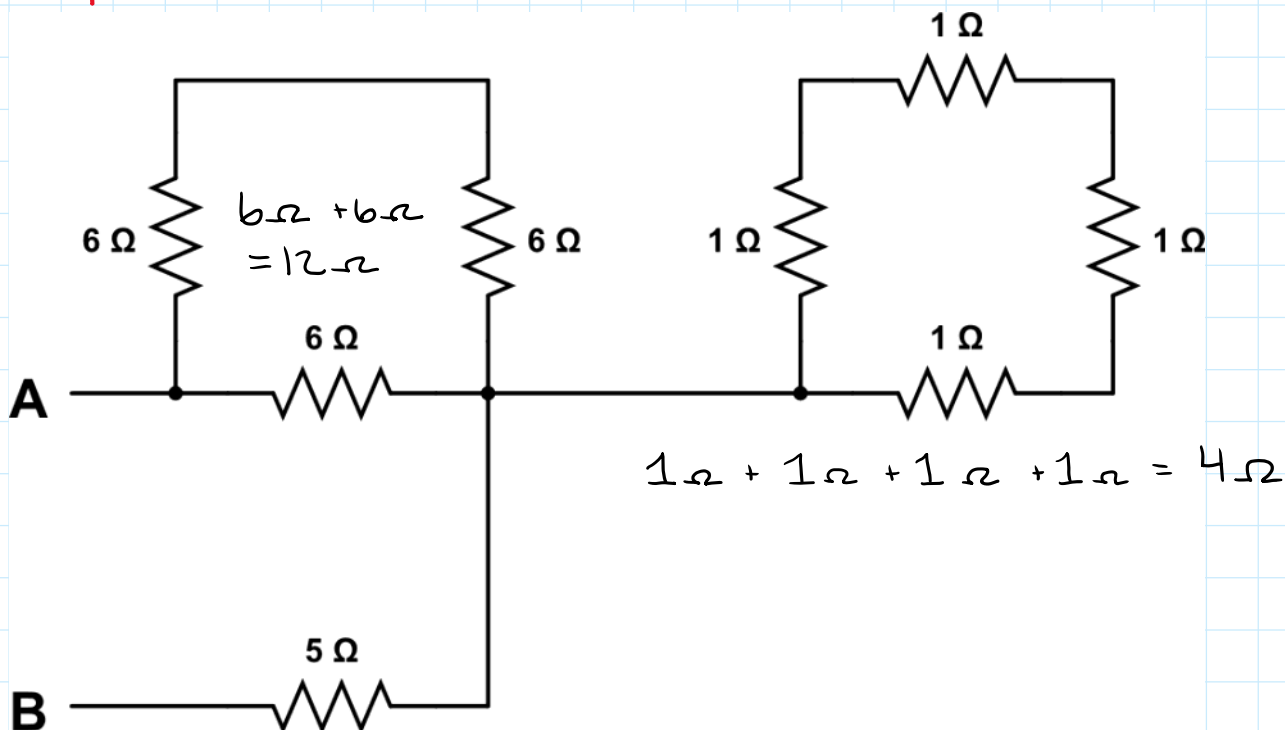


Step One: Identify short and open resistors

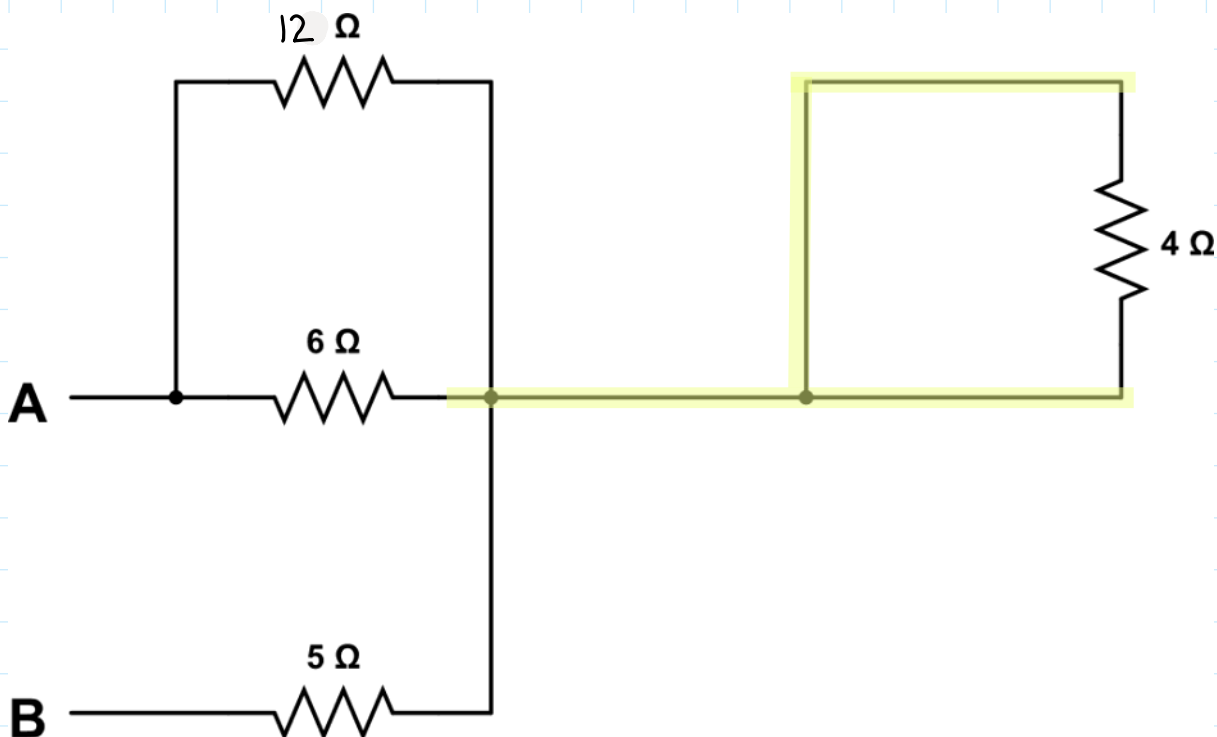


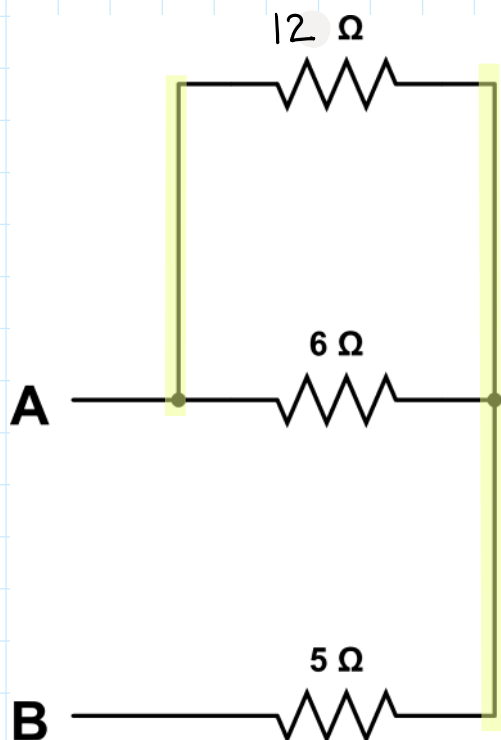
This $1\ \Omega$ resistor is shorted because both sides are connected to one node.

Step Two: Combine resistors in Series



After simplifying the right loop, we can see that both sides of the 4Ω resistor are connected to the same node, so that resistor becomes a short circuit.

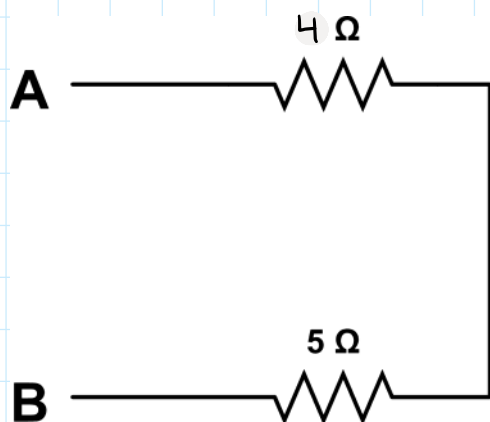




After highlighting our remaining nodes, we can see that both $6\ \Omega$ resistors are in parallel because they share 2 nodes

$$\frac{1}{\frac{1}{6} + \frac{1}{12}} = 4\ \Omega \text{ or}$$

$$\frac{6 \times 12}{6 + 12} = 4\ \Omega$$



Finally, we are left with a series circuit

$$4\ \Omega + 5\ \Omega = 9\ \Omega$$

The equivalent resistance between points A and B is $9\ \Omega$

$$R_{AB} = 9\ \Omega$$